$$\frac{1}{2} m v_0^2 n_0 \longrightarrow \frac{1}{2} m (n_0 + n_1)(v_0 + v_1)^2$$

$$[n_0 + n_1][v_0 + v_1]^2 - v_0^2 n_0$$

$$= n_0 v_0^2 + 2 n_0 v_0 v_1 + n_0 v_1^2 + n_1 v_0^2 + 2 n_1 v_0 v_1 + n_1 v_1^2 - v_0^2 n_0$$

$$= 2 n_0 v_0 v_1 + n_1 v_0^2 = v_0 (2 n_0 v_1 + n_1 v_0).$$

$$\frac{\partial n_1}{\partial t} + \nabla \cdot (n v) = 0.$$

$$\therefore \frac{\partial n_1}{\partial t} + \nabla \cdot [(n_0 + n_1)(v_0 + v_1)] = 0.$$

$$n_0 v_1 + n_1 v_0$$

$$\frac{\partial n_1}{\partial t} + n_0 \nabla \cdot \vec{v}_1 + \nabla n_1 \cdot \vec{v}_0 = 0.$$

$$i\omega n_1 = i k n_0 \vec{v}_1 + i k n_1 v_0 = 0.$$

$$\therefore \sim = v_0 n_0 v_1 = v_0 n_0 v_1 \cos(kx - \omega t).$$

$$-\frac{1}{\omega} v_0 n_0 v_1 \sin(kx - \omega t)$$